Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 1 Quiz 1 - Kinematics in One Dimension

Mock Exam F · Form 2: Advanced reasoning

Multiple Choice Questions: Select the best answer. (1 point each)

1. A cart starts at +9 m/s and maintains a = -1 m/s². Over T = v0/|a| + 2 s (approximately 11 s), what is the average speed for the entire interval?

(A) 7.727 m/s

(B) 1.932 m/s

(C) 3.864 m/s

(D) -3.864 m/s

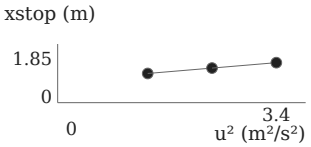
2. A cart brakes immediately at a constant magnitude of 2 m/s². Read its stopping coordinate along the track. Double braking deceleration from the same starting mark. Which X in the modified apparatus yields Y=1.3?

(A) 2.453

(B) 2.295

(C) 2.558

(D) 2.4



3. A cart is currently at x = -8 m with v = -17 m/s and a = 1 m/s². Which statement correctly describes the x–t slope, concavity, and change in speed?

(A) Slope is negative, concavity is up, and speed decreases.

(B) The object is currently moving to the right.

(C) The x–t slope is zero and the object is at rest.

(D) A negative velocity means a positive x–t slope.

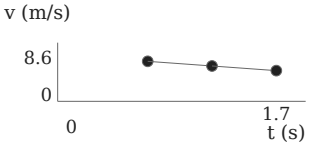
4. A bicycle moving right brakes uniformly. Use the model only until it stops. t (s) -> v (m/s): (0.7, 8.6); (1.2, 7.6); (1.7, 6.6). Both sensors multiply their axes by the same unknown positive gain k. What is the uncorrected slope?

(A) -0.4

(B) -2

(C) -1.2

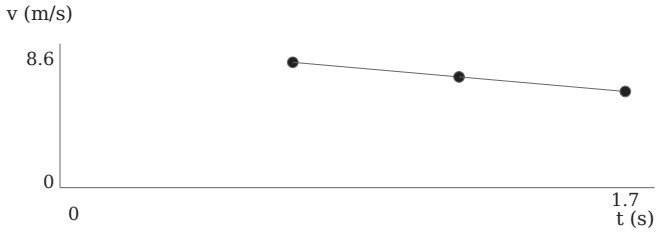
(D) -2.8



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A bicycle moving right brakes uniformly. Use the model only until it stops. t (s) -> v (m/s): (0.7, 8.6); (1.2, 7.6); (1.7, 6.6).

a. For the brake recorder setup, label the measured quantities and their reference directions or origins. **(1 pt)**



b. Double braking force at the same starting speed and mass. Predict the new slope and intercept and explain physically what changed. **(3 pt)**

6. A sensor records downward position for a weight dropped from rest. Use g=10 m/s². t² (s²) -> y (m): (1.4, 9); (2.4, 14); (3.4, 19).

a. Build the model for drop sensor from physics and predict both slope and intercept. **(2 pt)**

b. Release from the same coordinate where gravity is halved. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. A cart brakes immediately at a constant magnitude of 2 m/s². Read its stopping coordinate along the track. u² (m²/s²) -> xstop (m): (1.4, 1.35); (2.4, 1.6); (3.4, 1.85). Build the model for stopping-distance test from physics and predict both slope and intercept. **(3 pt)**